

Restaurant Sales Project

Phase 1 — Data Loading & Inspection

- How many rows and columns does the dataset contain?

```
>> (8010, 9)
```

- Which columns contain missing values?

```
>> item , price , quantity , order total , order date , payment  
method
```

- Which columns are numerical? Which are categorical?

```
>> Numerical ['Price', 'Quantity', 'Order Total']
```

```
>> Categorical ['Category', 'Item', 'Payment Method']
```

- Are any columns using an incorrect data type?

```
>> [Price , Quantity , Order date]
```

Phase 2 — Data Cleaning

- Which column has the most missing values?

```
>> Item
```

- How many missing values remain after cleaning?

```
>> Zero
```

- Which method did you use for each important missing-value problem, and why?

```
>> mapping – to fill item and price with the given price map
```

```
dropna() – to drop the price and item values that remained after the mapping
```

```
ffill() – to fill the date from the previous columns
```

- How many duplicate records were found and removed?

```
>> 4
```

- Which columns had incorrect data types?

```
>> [Price , Quantity , Order date]
```

- What should their final data types be?

>> [Price - float , Quantity - float , Order date - datetime]

- Were there invalid values that prevented conversion?

>> yes

- Which columns contained inconsistent values?

>> The main columns with inconsistent values were Item, Category, Price, Order Date, Payment Method, and Quantity.

- What inconsistencies did you find?

>> I found missing values and inconsistencies between related columns. For example, some Item and Category values were missing, while the corresponding Price or Item information was available. There were also missing values in Order Date, Payment Method, and Quantity.

- How did you standardize them?

>> I standardized the data by filling missing Item, Category, and Price values using relationships between the columns. I converted Order Date to a consistent datetime format and used forward fill (`ffill()`) for missing dates. Missing Payment Method values were filled based on the distribution of existing values, while missing Quantity values were replaced with the median.

- Is the dataset ready for analysis? Explain why.

>> Yes, the dataset is ready for analysis because the major data quality issues were addressed. Missing values were handled, inconsistent values were standardized, and the Order Date column was converted to the correct datetime format. These cleaning steps improved the accuracy and consistency of the data, making it suitable for analysis and visualization.

Phase 3 — Data Selection & Filtering

- Which orders have the highest order totals?

>> The highest order total is \$100. Several orders reached this value, including orders containing Steak.

- Which items have the highest quantities?

>> The items with the highest total quantities sold are:

1. Ice Cream — 1,022 units
2. Pasta Alfredo — 994 units
3. French Fries — 952 units
4. Water — 865 units

5. Side Salad — 802 units

- What does the sorted result reveal about restaurant sales?

>> The sorted results reveal that high-priced items, especially Steak, generate the highest order totals, while popular items such as Ice Cream and Pasta Alfredo are sold in the largest quantities. This shows that the restaurant has a mix of high-revenue premium items and high-demand popular items, both of which contribute to overall sales performance.

Phase 4 — Data Transformation

- What new columns did you create, and why are they useful?

>> Day Name: Extracted from the Order Date to identify the day of the week for each order. It is useful for analyzing sales patterns by day.

Customer Order Number: Used to identify how many orders each customer has placed. It helps distinguish between new and repeated customers.

Year-Month: Extracted from the Order Date to analyze sales trends and revenue over time.

- What business problem does each transformation help solve?

>> Day Name helps the restaurant understand which days have higher or lower sales, supporting better staffing and planning.

Customer Order Number helps identify repeat customers and customer purchasing behavior, which can support customer retention strategies.

Year-Month helps track monthly sales trends and identify changes or patterns over time.